

Shubham Rajpurohit

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M.Tech (IT) student at NIT Karnataka with strong foundations in systems programming, C/C++, and compiler concepts. Built low-level Unix systems tools, real-time ML inference pipelines, and production backend services — with measurable outcomes across academic and industry projects. Seeking a software engineering or systems internship to contribute to compilers, runtimes, or GPU computing infrastructure.

EDUCATION

National Institute of Technology Karnataka (NITK) M.Tech in Information Technology	Aug 2025 – July 2027 Mangalore, IN
Parul University B.Tech in Computer Science and Engineering <i>Relevant Coursework (M.Tech):</i> Advanced Data Structures & Algorithms, Distributed Systems, Web & Social Computing, Machine Learning, Deep Learning, Cybersecurity <i>(B.Tech):</i> Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering	Aug 2020 – July 2024 Vadodara, IN

EXPERIENCE

Walpar Nutritions Ltd. <i>Software Engineering Intern</i>	Ahmedabad, IN Dec 2023 – Feb 2024
- Designed and developed modular backend services for product, order, and transaction workflows using Python (Flask) and SQL, reducing manual processing overhead by 30% — delivering clean, maintainable code aligned with engineering best practices. - Optimised relational database schemas and query plans through indexing and query restructuring, reducing average API request latency by 25% and improving overall system responsiveness. - Debugged and resolved 10+ production issues across backend services and REST API endpoints through systematic root-cause analysis, and collaborated with product teams to deliver features within Agile/iterative sprint cycles.	

PROJECTS

Mini Shell — Unix Shell in C++	<i>C++17 · POSIX APIs · Linux</i> 2023
- Built a fully functional Unix shell in C++17 featuring a custom tokeniser and recursive-descent parser that handles multi-stage pipelines, I/O redirection (<, >, >>), background execution, and built-in commands (cd, pwd, exit) — directly demonstrating compiler front-end and language parsing concepts. - Managed OS-level process lifecycle using POSIX fork/execvp/waitpid, implemented signal handlers for SIGINT and SIGCHLD, and handled zombie reaping and file descriptor management — showcasing strong systems and machine architecture fundamentals.	
Facial Emotion Recognition — ConvNeXt-Tiny	<i>Python · PyTorch · ConvNeXt · OpenCV · Gradio</i> 2025
- Fine-tuned ConvNeXt-Tiny on FER2013 and RAF-DB datasets with custom classification heads, achieving 84.32% accuracy on RAF-DB and 70.64% on FER2013 across 7 emotion classes. - Implemented weak-class specific augmentation and Class-Balanced Focal Loss to tackle severe class imbalance, improving minority class (Fear, Disgust, Surprise) F1-scores by 8–9%. - Deployed a real-time Gradio web application supporting 3 model variants with live confidence visualisation, achieving <20ms inference latency (135 FPS) on GPU hardware via two-phase transfer learning with progressive backbone unfreezing.	
Social Distancing Monitor — Intel Unnati Industrial Training	<i>Python · OpenVINO · YOLO · OpenCV</i> 2023
- Built a real-time social distancing violation detector using Intel's OpenVINO toolkit with pre-trained person detection and re-identification models on live and recorded video streams. - Implemented centroid-based distance estimation between detected bounding boxes to flag proximity violations with color-coded overlays, achieving ~32–35 FPS at 27–31ms inference latency on Intel hardware.	

TECHNICAL SKILLS

Languages:	C++17, Python, SQL, JavaScript
Systems & OS:	POSIX APIs, process management (fork/exec/wait), signal handling, file descriptors, Linux
Backend & Web:	Flask, REST APIs, HTML, CSS, Bootstrap, Gradio, Agile/Scrum development cycles
AI/ML & Computer Vision:	PyTorch, TensorFlow/Keras, OpenCV, OpenVINO, ConvNeXt, NumPy, Pandas, scikit-learn
Tools & Platforms:	Git, GitHub, VS Code, Linux/WSL2, Google Colab, Azure (AI Fundamentals, Cloud Fundamentals)

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals (2023)
- Microsoft Certified: Azure Fundamentals (2023)